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DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING

DIGITAL SIGNAL PROCESSING (IPCC21EC42)

MODULE-4 IIR FILTER DESIGN

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MODULE-4 : IIR FILTER

DESIGN

The purpose of a filter is to enhance a wanted signal relative to unwanted signals, interference or noise.

An IIR filter is described using the difference equation,

$$y(n) = b_0 x(n) + b_1 x(n-1) + \dots + b_m x(n-m) - a_1 y(n-1) - \dots - a_N y(n-N)$$

or the transfer function,

$$H(z) = \frac{Y(z)}{X(z)} = \frac{b_0 + b_1 z^{-1} + \dots + b_m z^{-m}}{1 + a_1 z^{-1} + \dots + a_N z^{-N}}$$

where b_i and a_i are the $(m+1)$ numerator and N denominator coefficients, respectively, $Y(z)$ and $X(z)$ are the z -transform functions of the filter input $x(n)$ and filter output $y(n)$.

This type of transfer function generally yields an impulse response of infinite duration.

Hence they are commonly referred to as infinite impulse response (IIR) filters.

Some observations related to IIR :

- 1) The IIR filter output $y(n)$ depends not only on the current input $x(n)$ and past inputs $x(n-1), \dots$ but also on the past outputs $y(n-1), \dots$ [recursive terms]. Its transfer function is a ratio of the numerator polynomial over the denominator polynomial, and its impulse response has an infinite number of terms.
- 2) Since the transfer function has the denominator polynomial, and its roots yields the location of poles. The poles of a designed IIR filter must be inside the unit circle on the z -plane to ensure stability.
- 3) Compared with the finite impulse response (FIR) filter, the IIR offers a much smaller filter size (order, N). Hence, the filter operation requires a fewer number of computations, but the linear phase is not easily obtained.

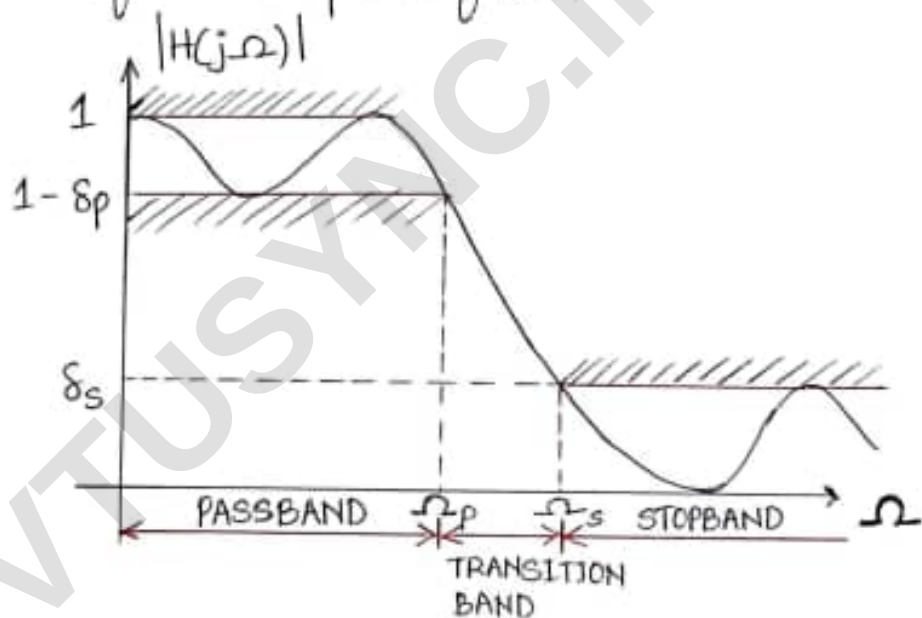
The IIR filter is preferred when a small size is called for but the application does not require a linear phase.

Analog Filter Specifications :

The frequency response describes how the filter reacts in the steady-state to sinusoidal inputs.

The definition of the frequency response specifications are hence important.

Figure shows a typical magnitude frequency response of a lowpass filter.



Ω_p and Ω_s demote the passband and stopband edge frequencies respectively.

The frequency region between Ω_p and Ω_s is called the transition band, no specifications are provided here.

Hatched areas in the passband and stopband indicate forbidden magnitude values.

The mathematical description of the frequency response as shown in figure, is

$$1 - \delta_p \leq |H(j\Omega)| \leq 1, \quad 0 \leq \Omega \leq \Omega_p$$

$$0 \leq |H(j\Omega)| \leq \delta_s, \quad |\Omega| \geq \Omega_s$$

The parameter δ_p is the tolerance of the magnitude response in the passband. The desired magnitude response in the passband is 1.

The parameter δ_s is the tolerance of the magnitude response in the stopband. The desired magnitude response in the stopband is 0.

$A_p = -20 \log(1 - \delta_p)$ is the passband ripple in dB.

$K_p = -A_p = 20 \log(1 - \delta_p)$ is the gain at $\Omega = \Omega_p$.

δ_s is called the stopband attenuation.

$A_s = -20 \log \delta_s$ is the stopband attenuation in dB.

$K_s = -A_s = 20 \log \delta_s$ is the gain at $\Omega = \Omega_s$.

The main classes of analog filters are Butterworth and Chebyshev filters. These filters differ in the nature of their magnitude responses as well as design and implementation.

A filter is said to be normalized if its cutoff frequency, Ω_c is 1 rad/sec.

BUTTERWORTH FILTERS

Butterworth filters have a very smooth passband, with a relatively wide transition region.

It has a magnitude response,

$$|H(j\Omega)| = \frac{1}{\left[1 + \left(\frac{\Omega}{\Omega_c}\right)^{2N}\right]^{1/2}}$$

N is the order of the filter and Ω_c is defined as the cutoff frequency, where the filter magnitude is $\frac{1}{\sqrt{2}}$ times the dc gain ($\Omega=0$).

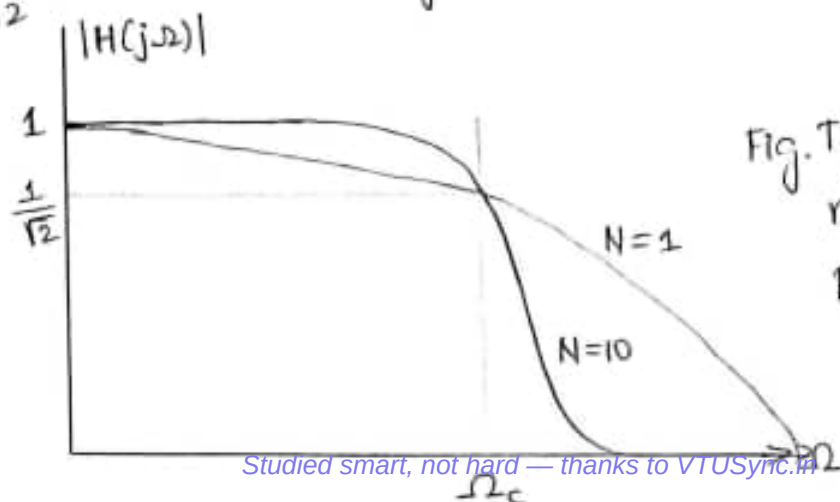


Fig. Typical magnitude responses of Butterworth filters

Following observations are made from fig above.

- 1) $|H(j\omega)| = 1$ for all N .
- 2) $|H(j\omega_c)| = \frac{1}{\sqrt{2}}$ at $\omega = \omega_c$ for all finite N .

$$\text{And } 20 \log_{10} |H(j\omega_c)| = -3.01 \text{ dB}$$

- 3) $|H(j\omega)| \rightarrow 0$ as $\omega \rightarrow \infty$.
- 4) Magnitude characteristics are said to be maximally flat i.e., $\left. \frac{d^n |H(j\omega)|}{d\omega^n} \right|_{\omega=0} = 0$ for $n=1, 2, \dots, 2N-1$.
- 5) $|H(j\omega)|$ is a monotonically decreasing function of frequency i.e., $|H(j\omega_2)| < |H(j\omega_1)|$ for any values of ω_1 and ω_2 where $0 \leq \omega_1 \leq \omega_2$.

The Poles of Butterworth:

The magnitude squared frequency response of the normalized ($\omega_c = 1$) lowpass Butterworth filter is

$$|H_N(j\omega)|^2 = \frac{1}{1 + \omega^{2N}}$$

$$H_N(j\omega) H_N(-j\omega) = \frac{1}{1 + \omega^{2N}}$$

Replace $j\omega$ by s and ω by $\frac{s}{j}$,

$$H_N(s) H_N(-s) = \frac{1}{1 + \left(\frac{s}{j}\right)^{2N}}$$

It has no finite zeros. The poles of $H_N(s) H_N(-s)$ is found by equating the denominator to zero,

$$1 + \left(\frac{s}{j}\right)^{2N} = 0$$

$$\Rightarrow s = (-1)^{1/2N} j$$

Since $-1 = e^{j\pi(2k+1)}$, $k=0,1,\dots$ and $j = e^{j\pi/2}$,

The poles are given by

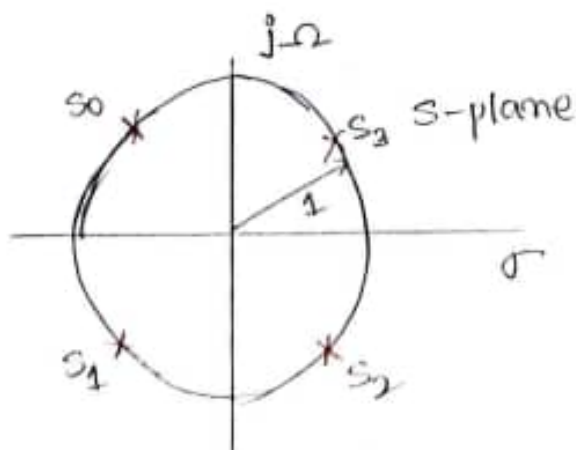
$$s_k = e^{j\pi \frac{(2k+1)}{2N}} e^{j\pi/2}, \quad k=0,1,\dots,2N-1$$

The poles lie on the unit circle ($\text{mag}=1$)

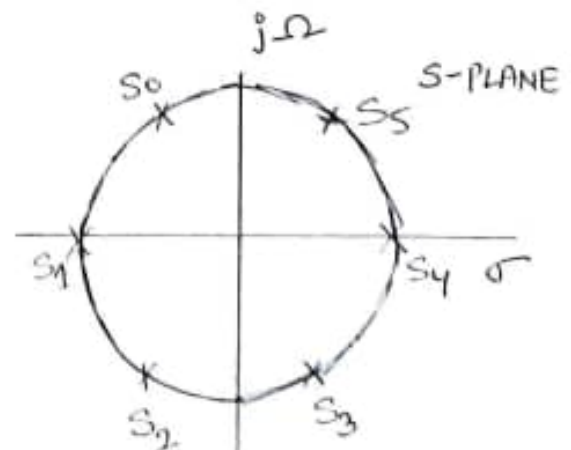
at angles given by

$$\theta_k = \frac{\pi k}{N} + \frac{\pi}{2N} + \frac{\pi}{2}, \quad k=0,1,\dots,2N-1$$

$$\text{and } s_k = 1 \angle \theta_k, \quad k=0,1,\dots,2N-1$$



POLES OF $H_N(s) H_N(-s)$, $N=2$



POLES OF $H_N(s) H_N(-s)$, $N=3$

The poles lie on a circle of unit radius and have a relative phase difference of $\frac{\pi}{N}$ radians.

The poles are distributed on the unit circle in the s-plane; half on the left-half plane, and half on the right-half plane. In order for $H_N(s)$ to be stable the left-half poles are assigned to $H_N(s)$.

$$H_N(s) = \frac{1}{\prod_{\text{LHP}} (s - s_k)} = \frac{1}{B_N(s)}$$

$s_k \rightarrow$ all the left-half poles of $H_N(s)H_N(-s)$.

$B_N(s) \rightarrow$ Butterworth polynomial of order N .

The first five Butterworth polynomials in real factored form.

Order N	Butterworth polynomial
1	$s+1$
2	$s^2 + \sqrt{2}s + 1$
3	$(s^2 + s + 1)(s+1)$
4	$(s^2 + 0.76536s + 1)(s^2 + 1.84776s + 1)$
5	$(s+1)(s^2 + 0.6180s + 1)(s^2 + 1.6180s + 1)$

To obtain Butterworth polynomial for $N=2$.

Left half poles of $H_N(s)$ are

$$s_0 = 1 \angle \theta_0 = 1 \angle \frac{3\pi}{4} = -\frac{1}{\sqrt{2}} + j\frac{1}{\sqrt{2}}$$

$$s_1 = 1 \angle \theta_1 = 1 \angle \frac{5\pi}{4} = -\frac{1}{\sqrt{2}} - j\frac{1}{\sqrt{2}}$$

$$B_2(s) = (s-s_0)(s-s_1) = \left(s + \frac{1}{\sqrt{2}} - j\frac{1}{\sqrt{2}}\right)\left(s + \frac{1}{\sqrt{2}} + j\frac{1}{\sqrt{2}}\right)$$

$$B_2(s) = \left(s + \frac{1}{\sqrt{2}}\right)^2 + \left(\frac{1}{\sqrt{2}}\right)^2 = s^2 + \sqrt{2}s + 1$$

Hence, the transfer function $H_2(s) = \frac{1}{s^2 + \sqrt{2}s + 1}$

Similarly for $N=3$, it can be shown

$$B_3(s) = (s^2 + s + 1)(s + 1)$$

$$\text{and } H_3(s) = \frac{1}{(s+1)(s^2+s+1)}$$

$$\text{by inserting } s_0 = 1 \angle \theta_0 = 1 \angle \frac{2\pi}{3} = -\frac{1}{2} + j\frac{\sqrt{3}}{2}$$

$$s_1 = 1 \angle \theta_1 = 1 \angle \pi = -1$$

$$s_2 = 1 \angle \theta_2 = 1 \angle \frac{4\pi}{3} = -\frac{1}{2} - j\frac{\sqrt{3}}{2}$$

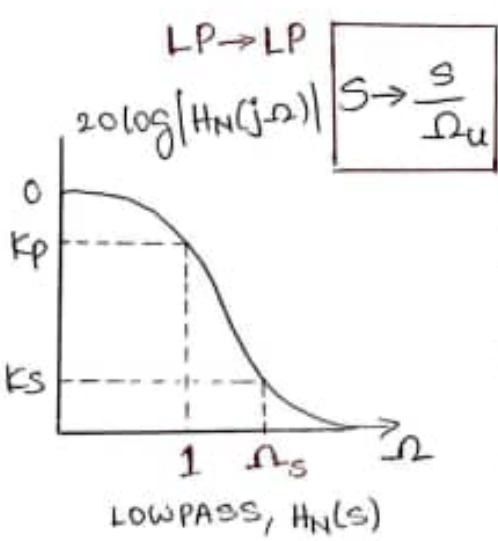
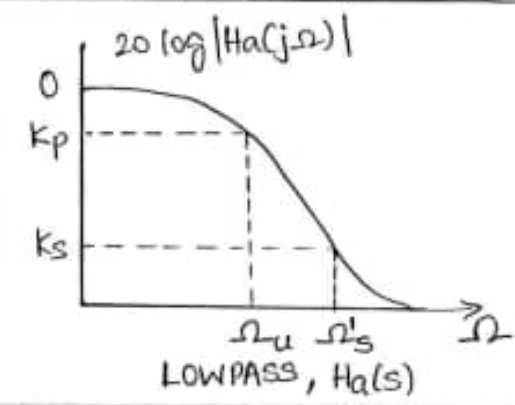
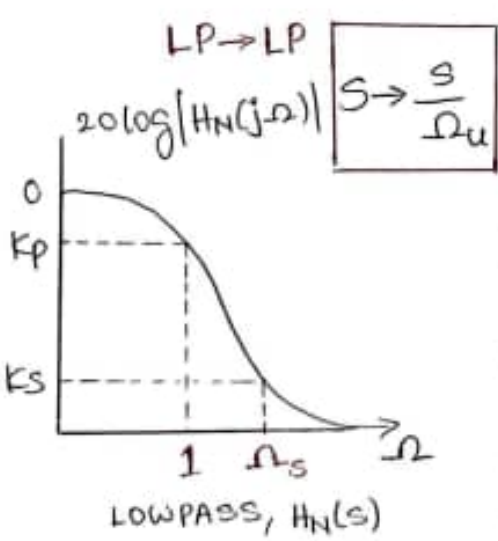
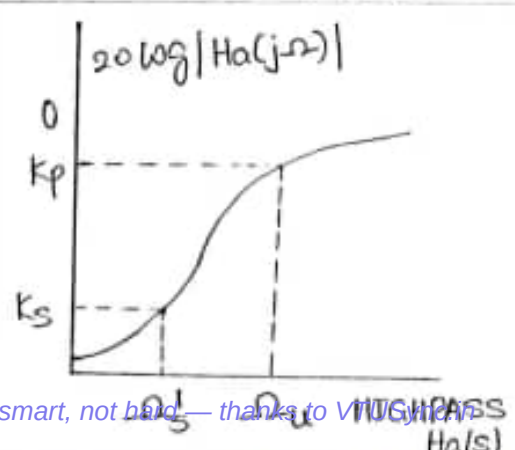
$$\text{im } B_3(s) = (s-s_0)(s-s_1)(s-s_2)$$

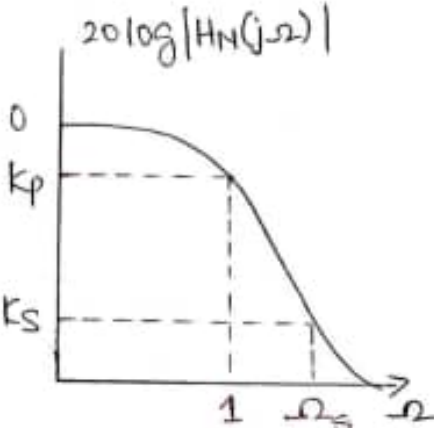
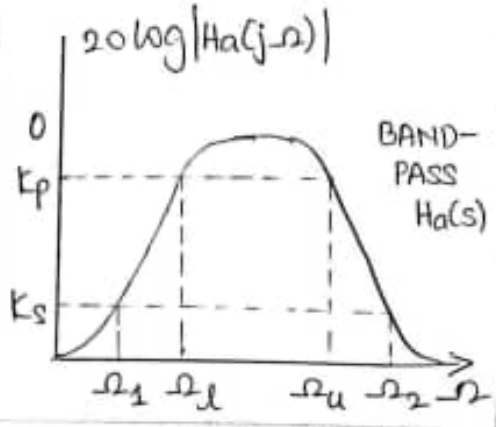
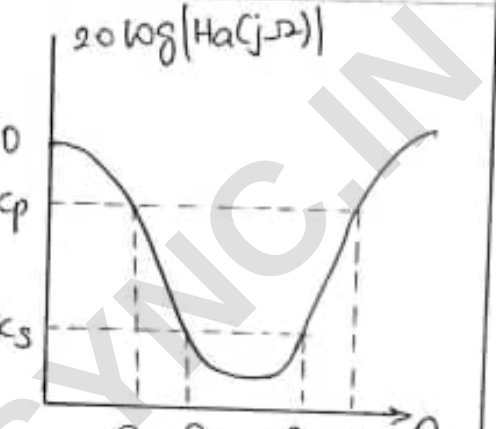
FREQUENCY TRANSFORMATIONS / SPECTRAL TRANSFORMATIONS

Let $H(s)$ denote the transfer function of a lowpass analog filter with a passband edge frequency $\Omega_p = 1$ rad/sec. This filter is known as analog lowpass normalised prototype.

If Ω_u is the desired passband edge frequency of the new lowpass filter, then the transfer function of the new lowpass filter is obtained using the transformation $s \rightarrow \frac{s}{\Omega_u}$. The cutoff frequency is translated from 1 rad/sec to Ω_u rad/sec.

ANALOG - TO - ANALOG TRANSFORMATIONS

PROTOTYPE FREQUENCY RESPONSE	TRANSFORMED FREQUENCY RESPONSE	BACKWARD DESIGN EQUATIONS
LP $\rightarrow$ LP  20 log H_N(j-Ω) $s \rightarrow \frac{s}{\Omega_u}$ LOWPASS, $H_N(s)$	 20 log H_a(j-Ω) LOWPASS, $H_a(s)$	$\Omega_s = \frac{\Omega'_s}{\Omega_u}$
LP $\rightarrow$ HP  20 log H_N(j-Ω) $s \rightarrow \frac{\Omega_u}{s}$ LOWPASS, $H_N(s)$	 20 log H_a(j-Ω) LOWPASS, $H_a(s)$	$\Omega_s = \frac{\Omega_u}{\Omega'_s}$

PROTOTYPE FREQUENCY RESPONSE	TRANSFORMED FREQUENCY RESPONSE	BACKWARD DESIGN EQUATIONS
LP $\rightarrow$ BP $s \rightarrow \frac{s^2 + \Omega_u \Omega_l}{s(\Omega_u - \Omega_l)}$  LOWPASS, $H_N(s)$	 BAND-PASS $H_a(s)$	$\Omega_s = \min \{ A , B \}$ $A = \frac{-\Omega_1^2 + \Omega_l \Omega_u}{\Omega_l (\Omega_u - \Omega_l)}$ $B = \frac{\Omega_2^2 - \Omega_l \Omega_u}{\Omega_2 (\Omega_u - \Omega_l)}$
LP $\rightarrow$ BS $s \rightarrow \frac{s(\Omega_u - \Omega_l)}{s^2 + \Omega_u \Omega_l}$	 BANDSTOP, $H_a(s)$	$\Omega_s = \min \{ A , B \}$ $A = \frac{\Omega_1 (\Omega_u - \Omega_l)}{-\Omega_1^2 + \Omega_u \Omega_l}$ $B = \frac{\Omega_2 (\Omega_u - \Omega_l)}{\Omega_2^2 - \Omega_l \Omega_u}$

Backward design equations are used to obtain the value Ω_s , the stopband edge frequency of the normalized lowpass filter. The normalized lowpass filter is designed first and then the suitable analog-to-analog transformation is applied.

Design of Lowpass Butterworth Filters

We are required to design a lowpass Butterworth filter to meet the following frequency-domain specifications.

$$K_p \leq 20 \log |H(j\Omega)| \leq 0 \quad \text{for all } \Omega \leq \Omega_p$$

$$20 \log |H(j\Omega)| \leq K_s \quad \text{for all } \Omega \geq \Omega_s$$

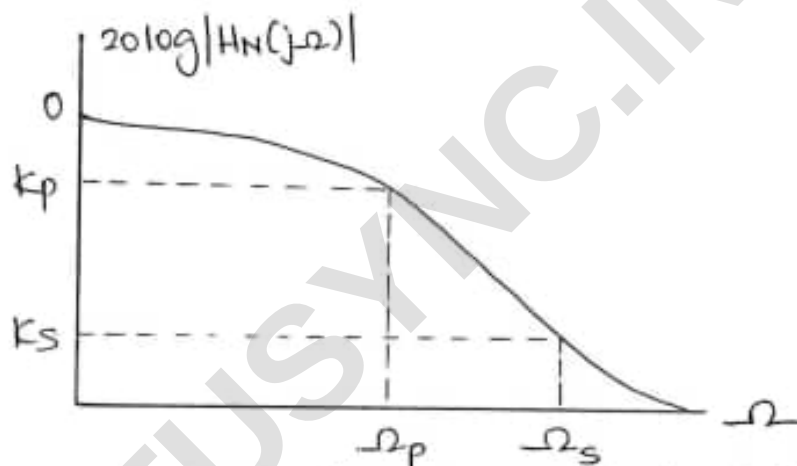


Fig. Freq. domain specifications of a low pass Butterworth filter.

The design of a lowpass filter involves determining its transfer function. The order N and the cutoff frequency Ω_c is necessary.

The magnitude response of a lowpass Butterworth filter is given by

$$|H(j\Omega)| = \frac{1}{\left[1 + \left(\frac{\Omega}{\Omega_c} \right)^{2N} \right]^{1/2}} \quad - (1)$$

Taking $20 \log$ on both sides of equation (1), we get

$$\begin{aligned} 20 \log |H(j\omega)| &= K = -20 \log \left[1 + \left(\frac{\omega}{\omega_c} \right)^{2N} \right]^{1/2} \\ &= -10 \log \left[1 + \left(\frac{\omega}{\omega_c} \right)^{2N} \right] \quad \text{--- (2)} \end{aligned}$$

From the figure, we note that at $\omega = \omega_p$, $K = K_p$.

Using this in equation (2),

$$\begin{aligned} K_p &= -10 \log \left[1 + \left(\frac{\omega_p}{\omega_c} \right)^{2N} \right] \\ \Rightarrow \left(\frac{\omega_p}{\omega_c} \right)^{2N} &= 10^{-K_p/10} - 1 \quad \text{--- (3)} \end{aligned}$$

Also, at $\omega = \omega_s$, $K = K_s$. Using this in eqn (2),

$$\begin{aligned} K_s &= -10 \log \left[1 + \left(\frac{\omega_s}{\omega_c} \right)^{2N} \right] \\ \Rightarrow \left(\frac{\omega_s}{\omega_c} \right)^{2N} &= 10^{-K_s/10} - 1 \quad \text{--- (4)} \end{aligned}$$

Dividing equation (3) by equation (4), we get,

$$\left(\frac{\omega_p}{\omega_s} \right)^{2N} = \frac{(10^{-K_p/10} - 1)}{(10^{-K_s/10} - 1)}$$

Taking log on both sides,

$$2N \log\left(\frac{\Omega_p}{\Omega_s}\right) = \log\left[\frac{(10^{-k_p/10} - 1)}{(10^{-k_s/10} - 1)}\right]$$

Filter
Order

$$N = \frac{\log\left[\frac{10^{-k_p/10} - 1}{10^{-k_s/10} - 1}\right]}{2 \log\left(\frac{\Omega_p}{\Omega_s}\right)}$$

It must be rounded up to the next larger integer value. Eg. If $N = 2.3$, take $N = 3$.

Once, the order N gets decided, the procedure for finding the cutoff frequency Ω_c is as follows.

- 1) If we desire to meet the passband requirement exactly and do better than the stopband requirement, then cutoff frequency Ω_c is selected from

Eqn (3) as

$$\Omega_c = \frac{\Omega_p}{(10^{-k_p/10} - 1)^{1/2N}}$$

- 2) If we wish to meet the requirement at Ω_s precisely and do better than our requirement at Ω_p , we find the cutoff frequency Ω_c from Eqn (4) as

$$\Omega_c = \frac{\Omega_s}{\left[10^{-K_s/10} - 1\right]^{1/2N}}$$

- 3) The third option is to take cutoff frequency as the arithmetic mean of the two cutoff frequencies found above. In this case, we do better in both the bands.

Numericals :

① A Butterworth lowpass filter has to meet the following specifications

- Passband gain, $K_p = -1$ dB at $\Omega_p = 4$ rad/sec
- stopband attenuation greater than or equal to 20 dB at $\Omega_s = 8$ rad/sec.

Determine the transfer function $H_a(s)$ of the lowest-order Butterworth filter to meet the above specifications.

Sol'n:-

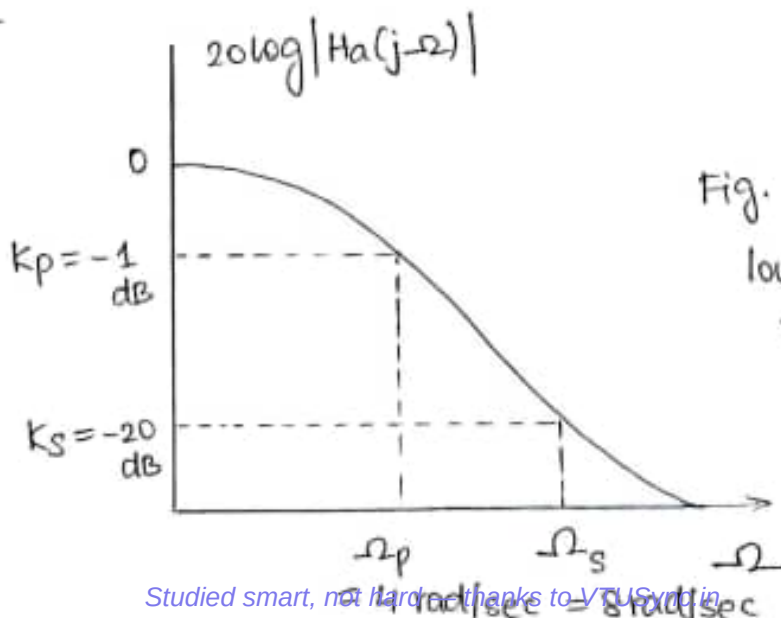


Fig. Specified lowpass magnitude response

The specified magnitude response of the lowpass Butterworth filter is shown in figure.

$$\text{Filter order, } N = \frac{\log \left[\frac{10^{-kp/10} - 1}{10^{-ks/10} - 1} \right]}{2 \log \left(\frac{\omega_p}{\omega_s} \right)}$$

$$N = 4.289$$

Rounding off to next largest integer, $N = 5$

To find the transfer function of 5th order normalized lowpass filter.

The poles of the normalized lowpass Butterworth filter are located at

$$s_k = 1 \angle \theta_k, \quad k = 0, 1, \dots, 2N-1$$

$$\text{where } \theta_k = \frac{\pi k}{N} + \frac{\pi}{2N} + \frac{\pi}{2}$$

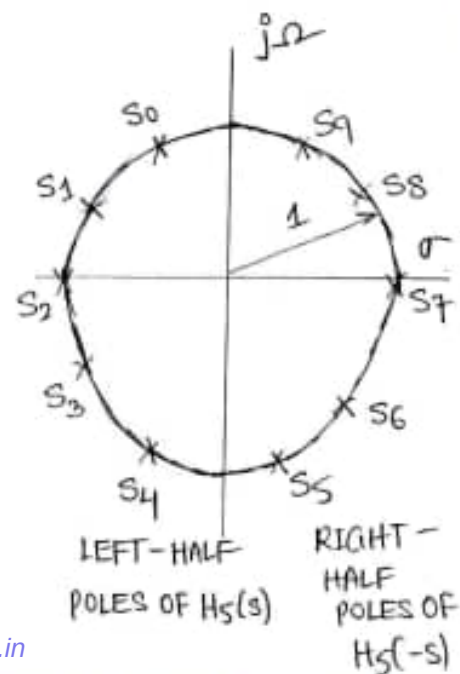
$$s_0 = 1 \angle \frac{6\pi}{10} = -0.309 + j0.951$$

$$s_1 = 1 \angle \frac{8\pi}{10} = -0.809 + j0.588$$

$$s_2 = 1 \angle \pi = -1$$

$$s_3 = 1 \angle \frac{12\pi}{10} = -0.809 - j0.588$$

$$s_4 = 1 \angle \frac{14\pi}{10} = -0.309 - j0.951$$



Left-Half poles of $H_s(s)$ considered for stability.

Transfer function of the s^{th} order normalized LPF is

$$H_s(s) = \frac{1}{\prod_{\text{LHP ONLY}} (s - s_k)} = \frac{1}{(s - s_0)(s - s_1)(s - s_2)(s - s_3)(s - s_4)}$$

$$\Rightarrow H_5(s) = \frac{1}{(s + 0.309 - j0.951)(s + 0.809 - j0.588)(s + 1)(s + 0.809 + j0.588)(s + 0.309 + j0.951)}$$

$$= \frac{1}{(s + 1) \left[(s + 0.309)^2 + (0.951)^2 \right] \left[(s + 0.809)^2 + (0.588)^2 \right]}$$

$$= \frac{1}{(s + 1) (s^2 + 0.6180s + 1) (s^2 + 1.6180s + 1)}$$

$$H_5(s) = \frac{1}{s^5 + 3.236s^4 + 5.236s^3 + 5.236s^2 + 3.236s + 1}$$

Find the cutoff frequency Ω_c to exactly meet the -1 dB requirement (passband) exactly and do better in the stopband.

$$\Omega_c = \frac{\Omega_p}{\left[10^{-K_p/10} - 1 \right]^{1/2N}} = 4.5787 \text{ rad/sec}$$

The specified lowpass filter is obtained by applying lowpass-to-lowpass transformation on the normalized lowpass filter.

$$\begin{aligned}
 H_a(s) &= H_s(s) \Big|_{s \rightarrow \frac{s}{\Omega_c}} \\
 &= H_s(s) \Big|_{s \rightarrow \frac{s}{4.5787}} \\
 &= \frac{1}{\left[\left(\frac{s}{4.5787} \right)^5 + 3.236 \left(\frac{s}{4.5787} \right)^4 + 5.236 \left(\frac{s}{4.5787} \right)^3 + 5.236 \left(\frac{s}{4.5787} \right)^2 + 3.236 \left(\frac{s}{4.5787} \right) + 1 \right]} \\
 H_a(s) &= \frac{2012.4}{s^5 + 14.82s^4 + 109.8s^3 + 502.6s^2 + 1422.3s + 2012.4}
 \end{aligned}$$

Verification of the design:-

The frequency response of an analog filter is obtained by letting $s=j\Omega$ in its transfer function.

$$H_a(j\Omega) = \frac{2012.4}{[(j\Omega)^5 + 14.82(j\Omega)^4 + 109.8(j\Omega)^3 + 502.6(j\Omega)^2 + 1422.3(j\Omega) + 2012.4]}$$
$$= \frac{2012.4}{(14.82\Omega^4 - 502.6\Omega^2 + 2012.4) + j(\Omega^5 - 109.8\Omega^3 + 1422.3\Omega)}$$

$$|H_a(j\Omega)| = \frac{2012.4}{\sqrt{(14.82\Omega^4 - 502.6\Omega^2 + 2012.4)^2 + (\Omega^5 - 109.8\Omega^3 + 1422.3\Omega)^2}}$$

Evaluated at $\Omega=4$ and $\Omega=8$.

$$20 \log |H_a(j\Omega)|_{\Omega=4} = -1 \text{ dB}$$

passband requirement exactly met

$$20 \log |H_a(j\Omega)|_{\Omega=8} = -24 \text{ dB}$$

stopband requirement bettered

② Let $H(s) = \frac{1}{s^2 + s + 1}$ represent the transfer function of a lowpass filter (not Butterworth) with a passband of 1 rad/sec. Use frequency transformation to find the transfer functions of the following analog filters.

- A lowpass filter with a passband of 10 rad/sec.
- A highpass filter with a cutoff frequency of 1 rad/sec.
- A highpass filter with a cutoff frequency of 10 rad/sec.
- A bandpass filter with a passband of 10 rad/sec and a center frequency of 100 rad/sec.
- A bandstop filter with a stopband of 2 rad/sec and a center frequency of 10 rad/sec.

Soln:-

Given $H(s) = \frac{1}{s^2 + s + 1}$

a) The lowpass-to-lowpass transformation is

$$s \rightarrow \frac{s}{\Omega_u}$$

Hence, the required lowpass filter is

$$H_a(s) = H(s) \Big|_{s \rightarrow \frac{s}{10}} = \frac{1}{\left(\frac{s}{10}\right)^2 + \left(\frac{s}{10}\right) + 1}$$

$$= \frac{100}{s^2 + 10s + 100}$$

b) The lowpass-to-highpass transformation is

$$s \rightarrow \frac{-\Omega_u}{s}$$

Hence, the required highpass filter is

$$H_a(s) = H(s) \Big|_{s \rightarrow \frac{1}{s}} = \frac{s^2}{s^2 + s + 1}$$

$$c) H_a(s) = \frac{1}{s^2 + s + 1} \Big|_{s \rightarrow \frac{10}{s}} = \frac{1}{\left(\frac{10}{s}\right)^2 + \left(\frac{10}{s}\right) + 1}$$

$$= \frac{s^2}{s^2 + 10s + 100}$$

d) The lowpass-to-bandpass transformation is

$$s \rightarrow \frac{s^2 + \Omega_u \Omega_l}{s(\Omega_u - \Omega_l)} = \frac{s^2 + \Omega_0^2}{s B_0}$$

where $\Omega_0 = \sqrt{\Omega_u \Omega_l}$ is the geometric center frequency and $B_0 = \Omega_u - \Omega_l$ is the passband bandwidth.

Hence, the required bandpass filter is

$$H_a(s) = \frac{1}{s^2 + s + 1} \Big|_{s \rightarrow \frac{s^2 + 10^4}{10s}}$$

$$H_a(s) = \frac{100 s^2}{s^4 + 10s^3 + 20100s^2 + 10^5 s + 10^8}$$

e) The lowpass-to-bandstop transformation is

$$s \rightarrow \frac{s(-\Omega_u - \Omega_l)}{s^2 + \Omega_u \Omega_l} = \frac{s B_0}{s^2 + \Omega_0^2}$$

Hence, the required bandstop filter is

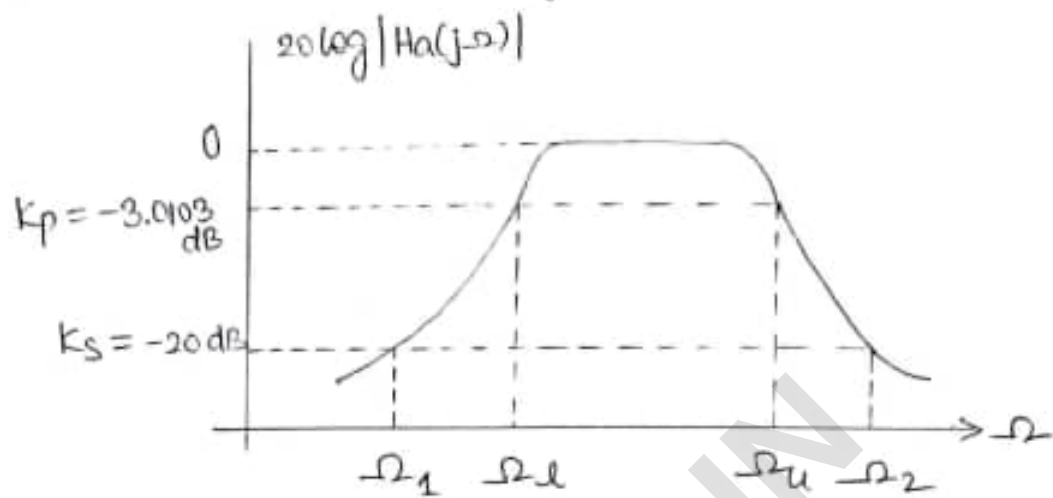
$$\begin{aligned} H_a(s) &= \frac{1}{s^2 + s + 1} \bigg|_{s \rightarrow \frac{2s}{s^2 + 100}} \\ &= \frac{(s^2 + 100)^2}{s^4 + 2s^3 + 2014s^2 + 200s + 10^4} \end{aligned}$$

③ Design an analog bandpass filter to meet the following frequency-domain specifications:

- a -3.0103 dB upper and lower cutoff frequency of 50 Hz and 20 kHz,
- a stopband attenuation of at least 20 dB at 20 Hz and 45 kHz, and
- a monotonic frequency response.

Soln:- A Butterworth filter is used to achieve monotonic frequency response.

The magnitude response of the specified bandpass filter is as shown in fig.



We have, $\Omega_1 = 2\pi \times 20 = 125.663 \text{ rad/sec}$

$$\Omega_2 = 2\pi \times 45 \times 10^3 = 2.827 \times 10^5 \text{ rad/sec}$$

$$\Omega_u = 2\pi \times 20 \times 10^3 = 1.257 \times 10^5 \text{ rad/sec}$$

$$\Omega_l = 2\pi \times 50 = 314.159 \text{ rad/sec}$$

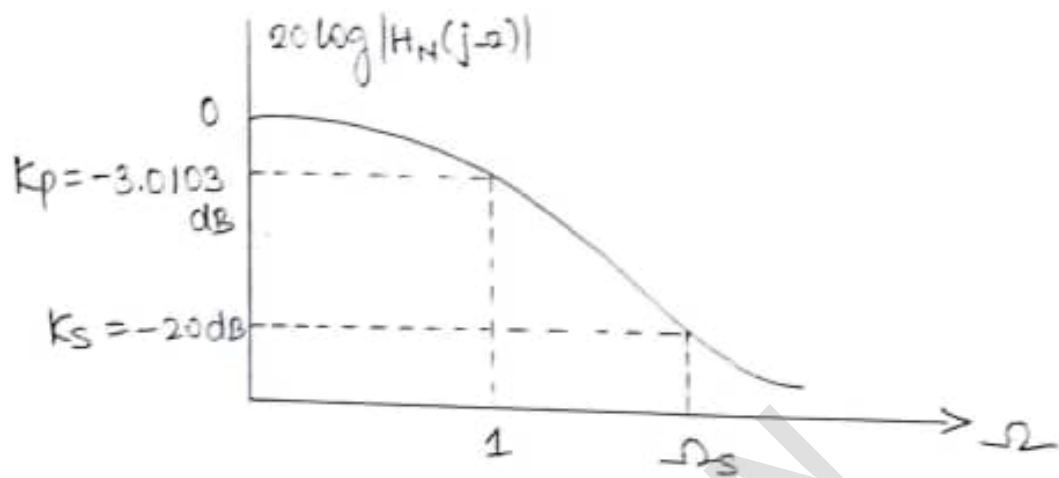
Use the backward design equation to find the stopband edge frequency Ω_s of the normalised lowpass filter.

$$A = \frac{-\Omega_1^2 + \Omega_l \Omega_u}{\Omega_1 (\Omega_u - \Omega_l)} = 2.51$$

$$B = \frac{\Omega_2^2 - \Omega_l \Omega_u}{\Omega_2 (\Omega_u - \Omega_l)} = 2.25$$

$$\Omega_s = \min \{ |A|, |B| \} = \underline{\underline{2.25}}$$

The magnitude response of the normalized lowpass filter is as shown below.



Design the normalized lowpass filter and then apply frequency transformation to convert to bandpass.

Order, N of lowpass Butterworth filter

$$N = \frac{\log \left[\frac{10^{-K_p/10} - 1}{10^{-K_s/10} - 1} \right]}{2 \log \left(\frac{1}{\Omega_s} \right)}$$

$$= \frac{\log \left[\frac{10^{0.30103} - 1}{10^2 - 1} \right]}{2 \log \left(\frac{1}{2.25} \right)} = 2.83$$

Rounding off to the next larger integer, $N=3$

The poles are given by :

$$s_k = 1 \angle \theta_k$$

$$= 1 \angle \left[\frac{\pi k}{N} + \frac{\pi}{2N} + \frac{\pi}{2} \right], \quad k=0, 1, \dots, 2N-1$$

For $N=3$, $s_k = 1 \angle \frac{\pi k}{3} + \frac{\pi}{6} + \frac{\pi}{2}$, $k=0,1,\dots,5$

Left half poles are obtained taking $k=0,1,2$.

k	s_k
0	$-\frac{1}{2} + j\frac{\sqrt{3}}{2}$
1	-1
2	$-\frac{1}{2} - j\frac{\sqrt{3}}{2}$

LHPs of $H_3(s)$

Hence, $H_3(s) = \frac{1}{(s-s_0)(s-s_1)(s-s_2)}$

$$= \frac{1}{\left(s + \frac{1}{2} - j\frac{\sqrt{3}}{2}\right)(s+1)\left(s + \frac{1}{2} + j\frac{\sqrt{3}}{2}\right)}$$

$$H_3(s) = \frac{1}{(s+1) \left[\left(s + \frac{1}{2}\right)^2 + \left(\frac{\sqrt{3}}{2}\right)^2 \right]} = \frac{1}{s^3 + 2s^2 + 2s + 1}$$

Applying lowpass-to-bandpass transformation on $H_3(s)$,

$$H_a(s) = H_3(s) \Big|_{s \rightarrow \frac{s^2 + \Omega_u \Omega_L}{s(\Omega_u - \Omega_L)}}$$

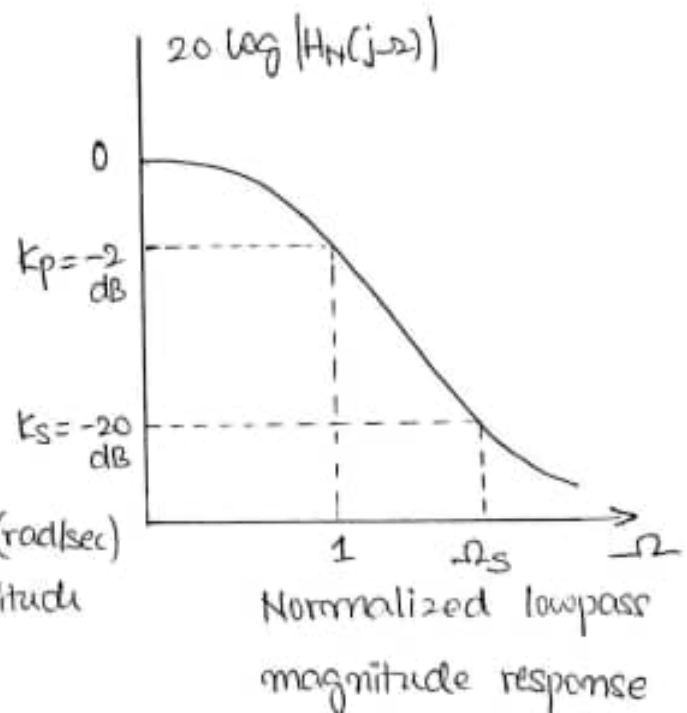
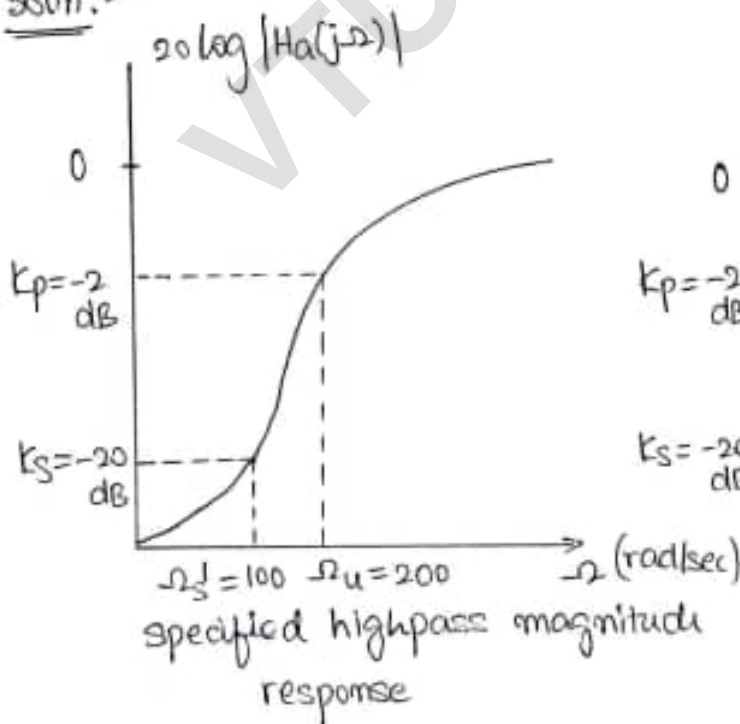
$$= H_3(s) \Big|_{s \rightarrow \frac{s^2 + 3.949 \times 10^7}{s(1.2538 \times 10^5)}}$$

$$H_a(s) = \frac{1.9695 \times 10^{15} s^3}{(s^6 + 2.51 \times 10^5 s^5 + 3.154 \times 10^{10} s^4 + 1.989 \times 10^{15} s^3 + 1.2453 \times 10^{18} s^2 + 3.9073 \times 10^{20} s + 6.1529 \times 10^{22})}$$

④ Design a Butterworth analog highpass filter that will meet the following specifications:

- Maximum passband attenuation = 2 dB
- Passband edge frequency = 200 rad/sec
- Minimum stopband attenuation = 20 dB
- Stopband edge frequency = 100 rad/sec

Soln:-



The stopband edge frequency of the normalized lowpass Butterworth filter $\omega_s = \frac{\omega_u}{\omega_s} = \frac{200}{100} = 2$

The normalized lowpass Butterworth filter now has the following frequency-domain specifications:

$$\Omega_p = 1, \quad K_p = -2 \text{ dB}$$

$$\Omega_s = 2, \quad K_s = -20 \text{ dB}$$

The order of the normalized Butterworth filter

$$N = \frac{\log \left[\frac{10^{-K_p/10} - 1}{10^{-K_s/10} - 1} \right]}{2 \log \left(\frac{\Omega_p}{\Omega_s} \right)}$$

$$= \frac{\log \left[(10^{0.2} - 1) / (10^2 - 1) \right]}{2 \log \left(\frac{1}{2} \right)}$$

$$= 3.7$$

Rounding off to the next larger integer, $N=4$.

Poles: $s_k = 1 \angle \theta_k = 1 \angle \left[\frac{k\pi}{N} + \frac{\pi}{2N} + \frac{\pi}{2} \right], \quad k=0, 1, \dots, 2N-1$

Here $N=4$, $s_k = 1 \angle \left[\frac{\pi k}{4} + \frac{\pi}{8} + \frac{\pi}{2} \right], \quad k=0, 1, \dots, 7$

k	s_k
-----	-------

0	$-0.3827 + j0.9239$
---	---------------------

1	$-0.9239 + j0.3827$
---	---------------------

2	$-0.9239 - j0.3827$
---	---------------------

3	$-0.3827 - j0.9239$
---	---------------------

LHP of $H_4(s)$

$$\text{Thus, } H_4(s) = \frac{1}{(s-s_0)(s-s_1)(s-s_2)(s-s_3)}$$

$$= \frac{1}{(s^2 + 0.76536s + 1)(s^2 + 1.84776s + 1)}$$

Let us determine the cutoff frequency Ω_c of the lowpass filter (prototype) so that -2dB requirement is satisfied exactly and do better in the stopband.

$$\Omega_c = \frac{\Omega_p}{\left(10^{-k_p/10} - 1\right)^{1/2N}} = \frac{1}{\left(10^{0.2} - 1\right)^{1/8}}$$

$$= 1.0693 \text{ rad/sec}$$

Hence, the lowpass filter prototype is

$$H_p(s) = H_4(s) \Big|_{s \rightarrow \frac{s}{\Omega_c}} = H_4(s) \Big|_{s \rightarrow \frac{s}{1.0693}}$$

To get the specified highpass filter $H_a(s)$, apply lowpass-to-highpass transformation on $H_p(s)$.

$$H_a(s) = H_p(s) \Big|_{s \rightarrow \frac{\Omega_u}{s}} = H_p(s) \Big|_{s \rightarrow \frac{200}{s}}$$

$$= H_4(s) \Big|_{s \rightarrow \frac{200}{1.0693s}} = H_4(s) \Big|_{s \rightarrow \frac{187.031}{s}}$$

$$H_a(s) = \frac{s^4}{(s^2 + 143.1464s + 34980.7521)(s^2 + 345.5892s + 34980.7521)}$$

- i) Sketch direct form-II, parallel and cascade realization of digital filters with transfer function

$$H(z) = \frac{1 - \frac{1}{5} z^{-1}}{\left(1 - \frac{1}{2} z^{-1} + \frac{1}{3} z^{-2}\right) \left(1 + \frac{1}{4} z^{-1}\right)}$$

- i) Direct Form - I

$$H(z) = \frac{1 - \frac{1}{5} z^{-1}}{1 + \frac{1}{4} z^{-1} - \frac{1}{2} z^{-1} - \frac{1}{8} z^{-2} + \frac{1}{3} z^{-2} + \frac{1}{12} z^{-3}}$$

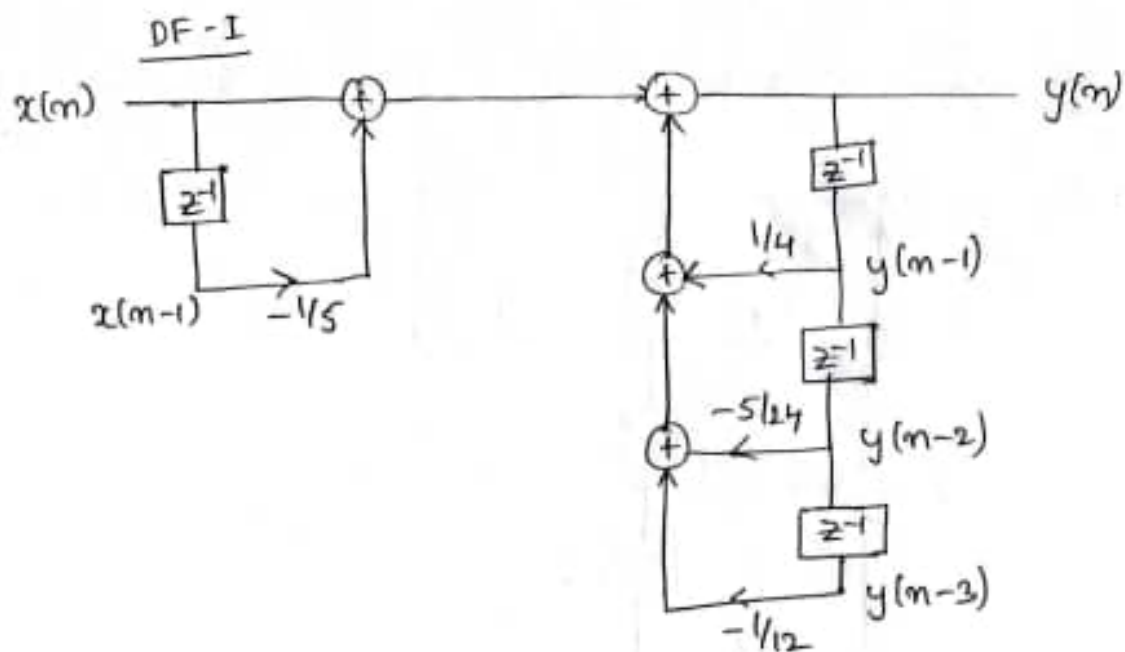
$$H(z) = \frac{Y(z)}{X(z)} = \frac{1 - \frac{1}{5} z^{-1}}{1 - \frac{1}{4} z^{-1} + \frac{5}{24} z^{-2} + \frac{1}{12} z^{-3}}$$

$$\therefore Y(z) \left[1 - \frac{1}{4} z^{-1} + \frac{5}{24} z^{-2} + \frac{1}{12} z^{-3} \right] = X(z) \left[1 - \frac{1}{5} z^{-1} \right]$$

Taking IZT,

$$\begin{aligned} y(n) - \frac{1}{4} y(n-1) + \frac{5}{24} y(n-2) + \frac{1}{12} y(n-3) \\ = x(n) - \frac{1}{5} x(n-1) \end{aligned}$$

$$\boxed{y(n) = x(n) - \frac{1}{5} x(n-1) + \frac{1}{4} y(n-1) - \frac{5}{24} y(n-2) - \frac{1}{12} y(n-3)}$$



ii) Direct Form II

$$H(z) = \frac{1 - \frac{1}{5}z^{-1}}{1 - \frac{1}{4}z^{-1} + \frac{5}{24}z^{-2} + \frac{1}{12}z^{-3}}$$

$$\text{Let } H(z) = \frac{Y(z)}{X(z)} = \frac{V(z)}{V(z)} = \frac{Y(z)}{V(z)} \cdot \frac{V(z)}{X(z)}$$

$$\text{Let } \frac{Y(z)}{V(z)} = 1 - \frac{1}{5}z^{-1} \xrightarrow{\text{IDT}} \boxed{y(n) = v(n) - \frac{1}{5}v(n-1)} \quad \text{L(a)}$$

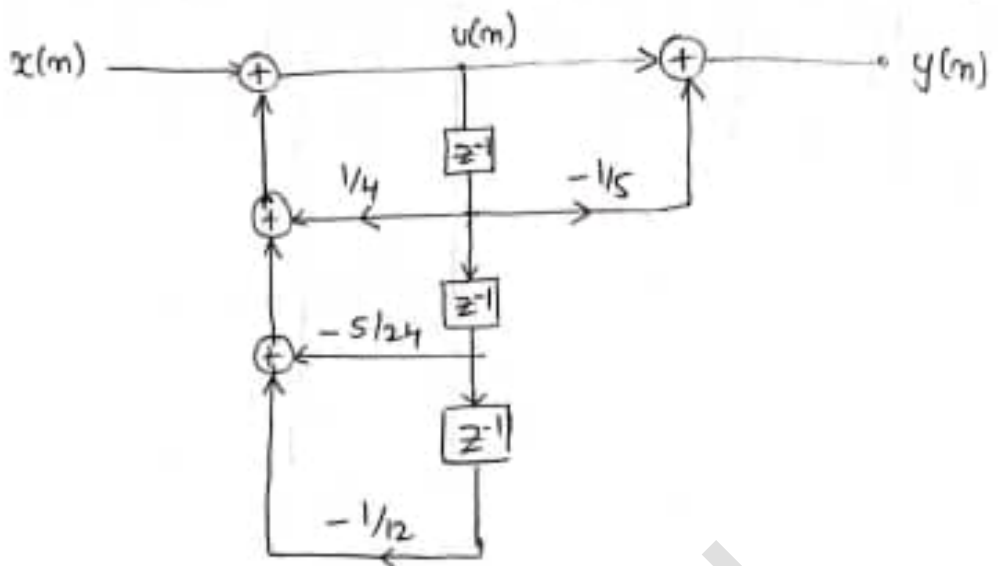
$$\Delta \frac{V(z)}{X(z)} = \frac{1}{1 - \frac{1}{4}z^{-1} + \frac{5}{24}z^{-2} + \frac{1}{12}z^{-3}}$$

Taking IDT,

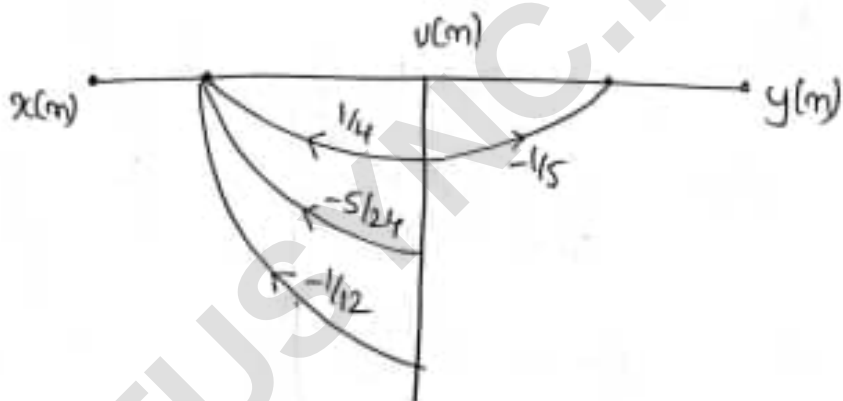
$$v(n) = \frac{1}{4}v(n-1) + \frac{5}{24}v(n-2) + \frac{1}{12}v(n-3) = x(n)$$

$$\text{or } \boxed{v(n) = x(n) + \frac{1}{4}v(n-1) - \frac{5}{24}v(n-2) - \frac{1}{12}v(n-3)} \quad \text{L(b)}$$

From eqns (a) & (b), Direct Form-II Realization



Signal flow graph

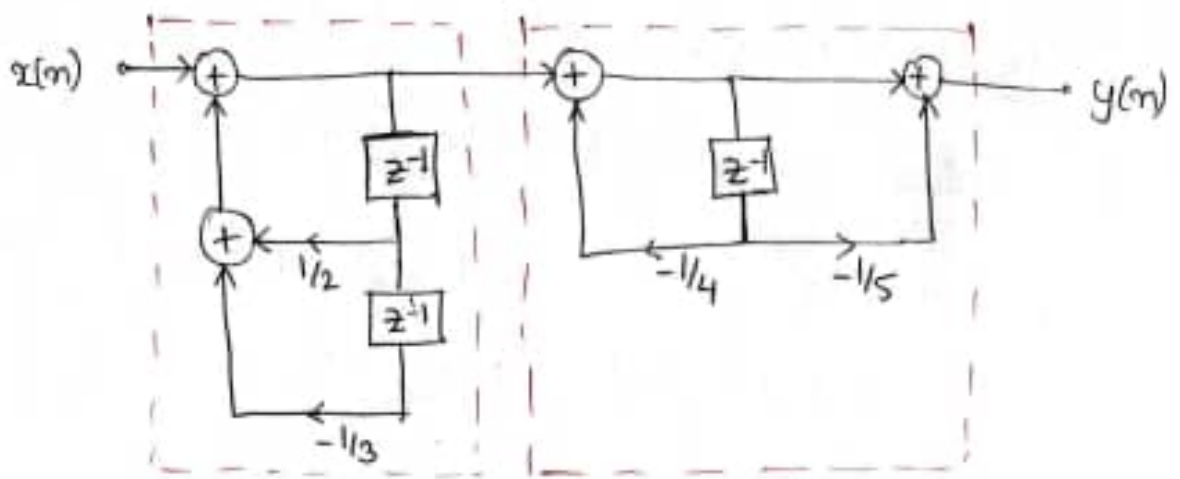


iii) Cascaded Form:-

$$H(z) = H_1(z) H_2(z)$$

$$H(z) = \frac{1 - 1/5 z^{-1}}{(1 - 1/2 z^{-1} + 1/3 z^{-2}) (1 + 1/4 z^{-1})}$$

$$H(z) = \frac{1}{\left(1 - \frac{1}{2} z^{-1} + \frac{1}{3} z^{-2}\right)} \times \frac{\left(1 - \frac{1}{5} z^{-1}\right)}{\left(1 + \frac{1}{4} z^{-1}\right)}$$



iv) Parallel Form:-

Consider $H(z) = \frac{1 - 1/5 z^{-1}}{(1 - 1/2 z^{-1} + 1/3 z^{-2})(1 + 1/4 z^{-1})}$

Replace $z^{-1} = p$

$$\therefore H(z) = \frac{1 - 1/5 p}{(1 - 1/2 p + 1/3 p^2)(1 + 1/4 p)}$$

Since the power of the numerator is less than the power of denominator we can perform partial fraction expansion.

$$\frac{1 - 1/5 p}{(1 - 1/2 p + 1/3 p^2)(1 + 1/4 p)} = \frac{A}{(1 + 1/4 p)} + \frac{Bp + C}{(1 - 1/2 p + 1/3 p^2)}$$

Solving for A,

$$A = \frac{1 - 1/5 p}{(1 - 1/2 p + 1/3 p^2)(1 + 1/4 p)} \cdot (1 + 1/4 p)$$

$$A = \frac{27}{125}$$

$$1 + \frac{1}{4}p = 0 \\ \text{or } p = -4$$

To find B & C:-

using co-efficient comparison method

$$\frac{(1 - 1/5 P)}{(1 - 1/2 P + 1/3 P^2)(1 + 1/4 P)} = \frac{A(1 - 1/2 P + 1/3 P^2) + (BP + C)(1 + 1/4 P)}{(1 - 1/2 P + 1/3 P^2)(1 + 1/4 P)}$$

Equating the numerators,

$$1 - 1/5 P = A - \frac{1}{2}AP + \frac{1}{3}AP^2 + BP + \frac{1}{4}BP^2 + C + \frac{1}{4}CP$$

Comparing the constant terms

$$1 = A + C$$

$$C = 1 - A = 1 - \frac{27}{125}$$

$$C = \frac{98}{125}$$

Comparing the coefficients of P,

$$-1/5 = -1/2 A + B + 1/4 C$$

$$B = -\frac{1}{5} + \frac{A}{2} - \frac{C}{4}$$

$$B = -\frac{36}{125}$$

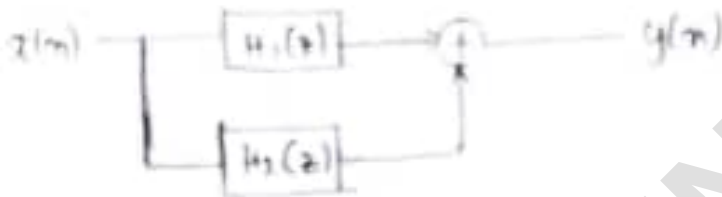
$$\therefore H(z) = \frac{27/125}{(1 + 1/4 P)} + \frac{\left(-\frac{36}{125} P + \frac{98}{125}\right)}{\left(1 - \frac{1}{2} P + \frac{1}{3} P^2\right)}$$

Replacing $p = z^{-1}$

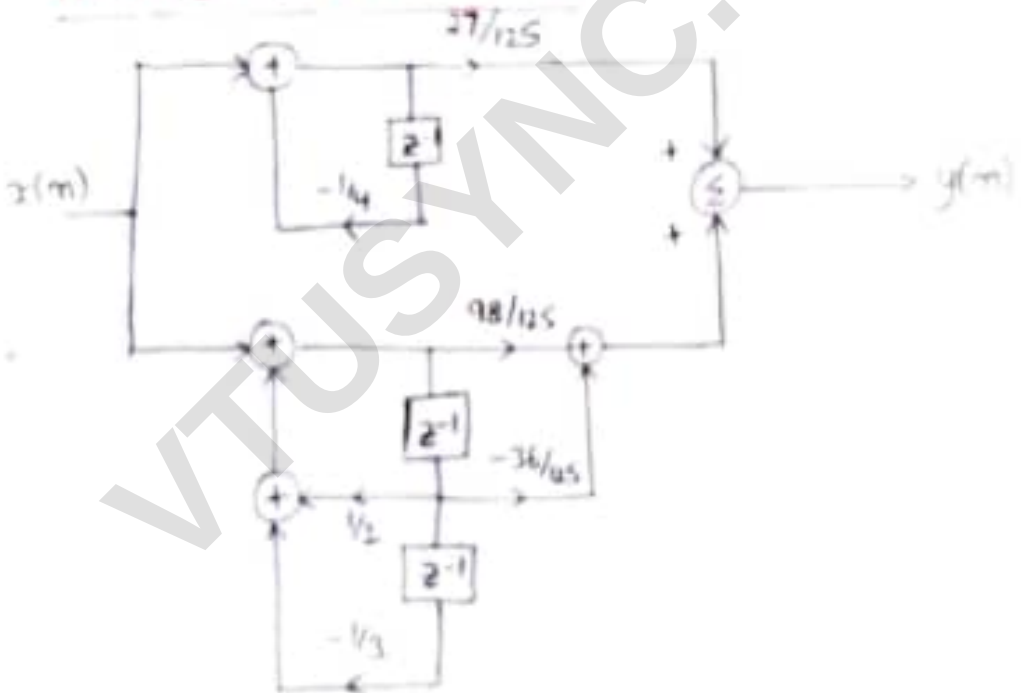
$$H(z) = \frac{22/125}{\left(1 + \frac{1}{4}z^{-1}\right)} \cdot \frac{\left(-\frac{36}{125}z^{-1} + \frac{98}{125}\right)}{\left(1 - \frac{1}{2}z^{-1} + \frac{1}{3}z^{-2}\right)}$$

$H_1(z)$

$H_2(z)$



PARALLEL FORM REALIZATION



2) Sketch the structures for $H(z)$

$$H(z) = \frac{8z^3 - 4z^2 + 11z - 2}{(z - 1/4)(z^2 - z + 1/2)}$$

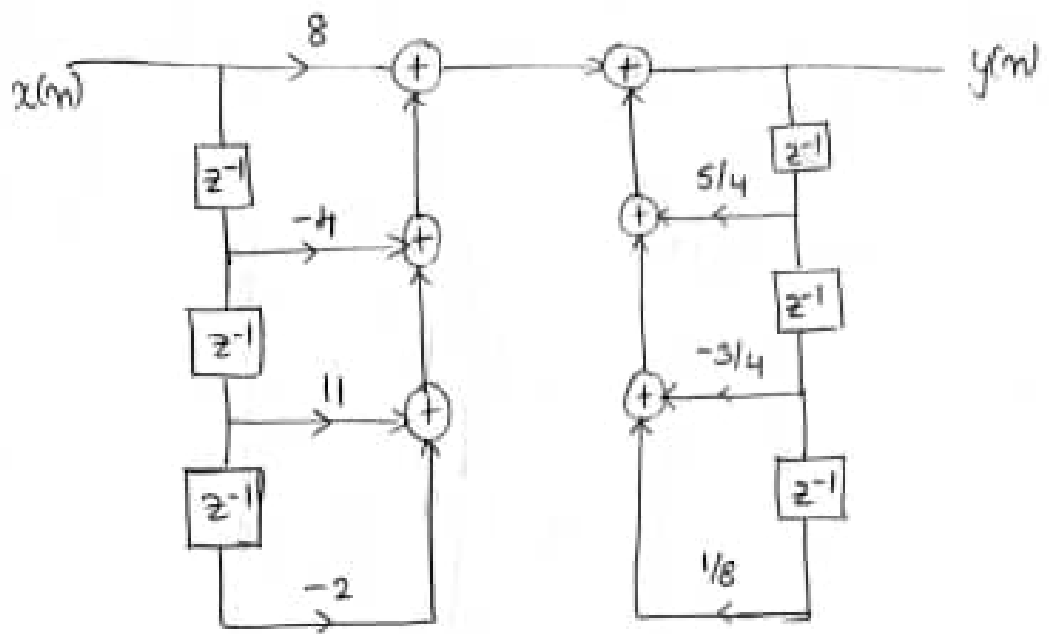
9) Direct Form - I

$$\begin{aligned} H(z) &= \frac{z^3 (8 - 4z^{-1} + 11z^{-2} - 2z^{-3})}{z \cdot z^2 (1 - \frac{1}{4}z^{-1})(1 - z^{-1} + \frac{1}{2}z^{-2})} \\ &= \frac{8 - 4z^{-1} + 11z^{-2} - 2z^{-3}}{1 - \frac{1}{4}z^{-1} - z^{-1} + \frac{1}{4}z^{-2} + \frac{1}{2}z^{-2} - \frac{1}{8}z^{-3}} \end{aligned}$$

$$H(z) = \frac{8 - 4z^{-1} + 11z^{-2} - 2z^{-3}}{1 - \frac{5}{4}z^{-1} + \frac{3}{4}z^{-2} - \frac{1}{8}z^{-3}}$$

$$\begin{aligned} \therefore y(m) &= \frac{5}{4} y(m-1) + \frac{3}{4} y(m-2) - \frac{1}{8} y(m-3) \\ &= 8x(m) - 4x(m-1) + 11x(m-2) - 2x(m-3) \end{aligned}$$

$$\begin{aligned} y(m) &= 8x(m) - 4x(m-1) + 11x(m-2) - 2x(m-3) \\ &\quad + \frac{5}{4} y(m-1) - \frac{3}{4} y(m-2) + \frac{1}{8} y(m-3) \end{aligned}$$



ii) Direct Form II

$$H(z) = \frac{Y(z)}{X(z)} \cdot \frac{V(z)}{V(z)} = \frac{Y(z)}{V(z)} \cdot \frac{V(z)}{X(z)}$$

$$\text{Let } \frac{Y(z)}{V(z)} = 8 - 4z^{-1} + 11z^{-2} - 2z^{-3}$$

$$\therefore y(n) = 8u(n) - 4u(n-1) + 11u(n-2) - 2u(n-3)$$

⌒ @

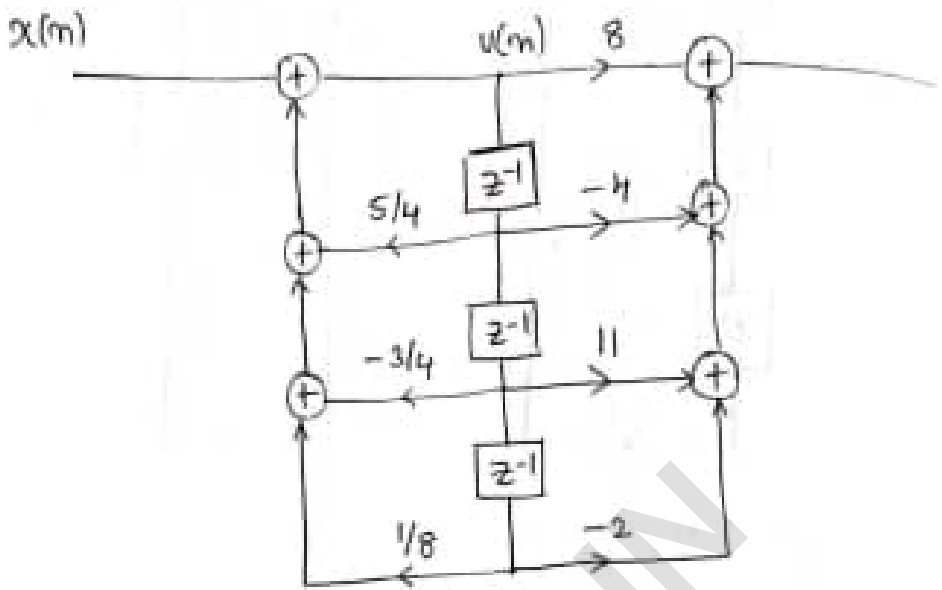
$$\therefore \frac{U(z)}{X(z)} = \frac{1}{1 - \frac{5}{4}z^{-1} + \frac{3}{4}z^{-2} - \frac{1}{8}z^{-3}}$$

$$x(n) = u(n) - \frac{5}{4}u(n-1) + \frac{3}{4}u(n-2) - \frac{1}{8}u(n-3)$$

$$u(n) = x(n) + \frac{5}{4}u(n-1) - \frac{3}{4}u(n-2) + \frac{1}{8}u(n-3)$$

⌒ (b)

Using eqn (a) & (b),



iii) Cascade Realization

$$H(z) = \frac{8z^3 - 4z^2 + 11z - 2}{(z - 1/4)(z^2 - z + 1/2)}$$

To convert numerator polynomial into factored form

consider

$$8z^3 - 4z^2 + 11z - 2 = 0$$

using calculator find the

first root

$$z = 0.1899$$

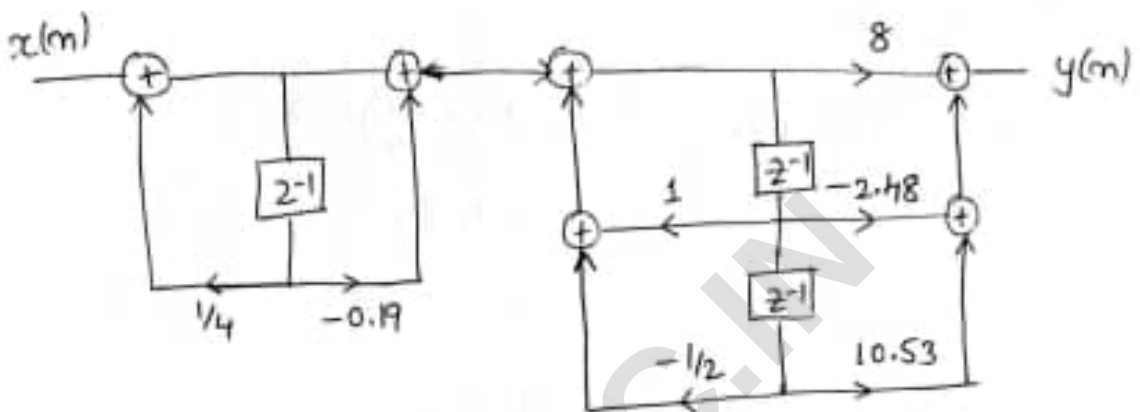
$$\therefore \boxed{z = 0.19}$$

0.19	8	-4	11	-2
	0	1.52	-0.47	2
	8	-2.48	10.53	0

$$\therefore 8z^3 - 4z^2 + 11z - 2 = (z - 0.19)(8z^2 - 2.48z + 10.53)$$

$$\therefore H(z) = \frac{(z - 0.19)(8z^2 - 2.48z + 10.53)}{(z - 1/4)(z^2 - z + 1/2)}$$

$$H(z) = \frac{(1 - 0.19z^{-1})}{(1 - 1/4z^{-1})} \left(\frac{8 - 2.48z^{-1} + 10.53z^{-2}}{1 - z^{-1} + 1/2z^{-2}} \right)$$



iv) Parallel Form: —

$$H(z) = \frac{8z^3 - 4z^2 + 11z - 2}{(z - 1/4)(z^2 - z + 1/2)}$$

To perform partial fraction expansion numerator power must be less compared to the denominator.

Divide both sides by z

$$\frac{H(z)}{z} = \frac{8z^3 - 4z^2 + 11z - 2}{z(z - 1/4)(z^2 - z + 1/2)}$$

$$\frac{H(z)}{z} = \frac{A}{z} + \frac{B}{z - 1/4} + \frac{Cz + D}{(z^2 - z + 1/2)}$$

$$A = \frac{8z^3 - 4z^2 + 11z - 2}{(z - 1/4)(z^2 - z + 1/2)} \Big|_{z=0}$$

↳ Gives complex factors

$$A = 16$$

$$B = \frac{8z^3 - 4z^2 + 11z - 2}{z(z^2 - z + \frac{1}{2})} \Big|_{z = \frac{1}{4}}$$

$$B = 8$$

To find C & D using coefficient comparison method.

$$\begin{aligned} 8z^3 - 4z^2 + 11z - 2 &= A(z - \frac{1}{4})(z^2 - z + \frac{1}{2}) \\ &+ B(z)(z^2 - z + \frac{1}{2}) \\ &+ (Cz + D)(z)(z - \frac{1}{4}) \end{aligned}$$

Equating coefficients of z^3 ,

$$8 = A + B + C$$

$$C = 8 - 16 - 8 = -16$$

$$C = -16$$

Equating coefficients of z^2 ,

$$-4 = -A - \frac{A}{4} - B - \frac{C}{4} + D$$

$$-4 = -16 - \frac{16}{4} - 8 + \frac{16}{4} + D$$

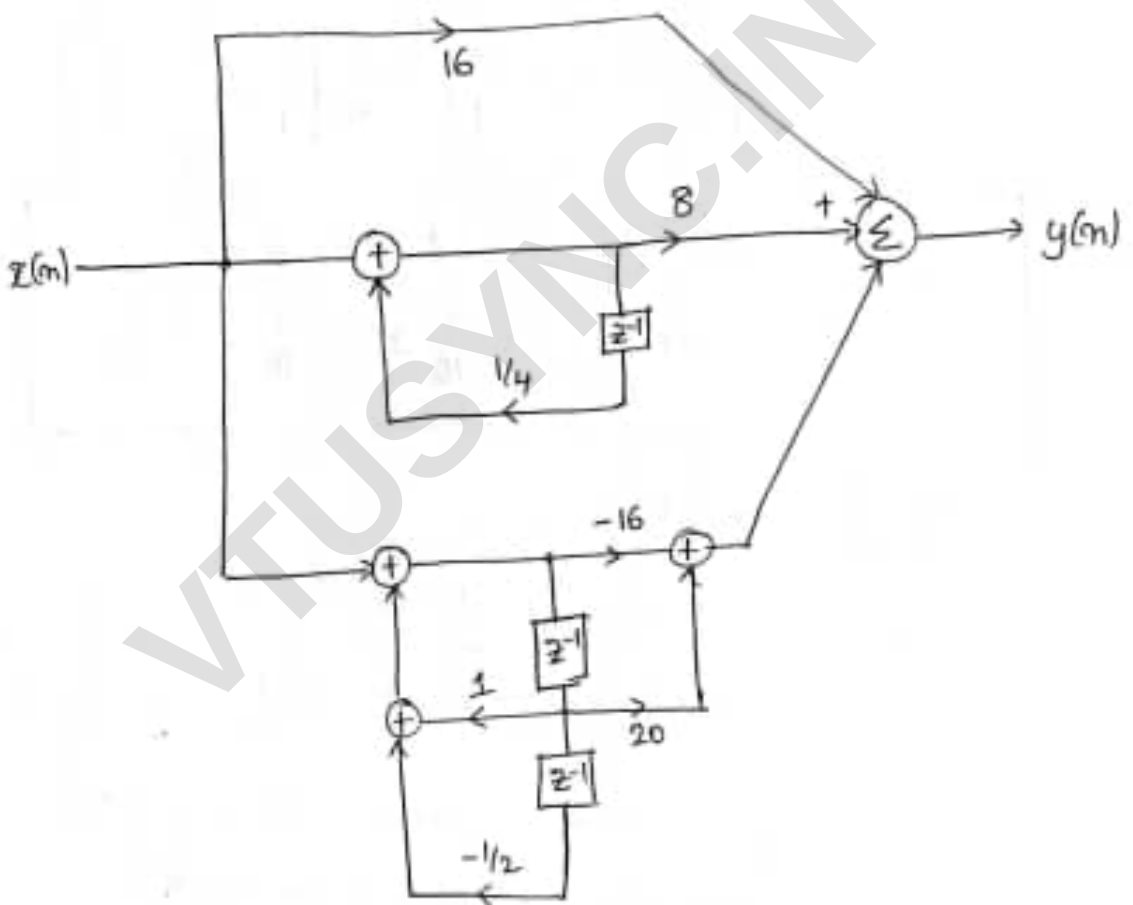
$$D = -4 + 16 + 8$$

$$\therefore D = 20$$

$$\frac{H(z)}{z} = \frac{16}{z} + \frac{8}{z - \frac{1}{4}} + \frac{-16z + 20}{(z^2 - z + \frac{1}{2})}$$

$$H(z) = 16 + \frac{8z}{z - \frac{1}{4}} + \frac{-16z^2 + 20z}{z^2 - z + \frac{1}{2}}$$

$$H(z) = \underbrace{16}_{H_1(z)} + \underbrace{\frac{8}{1 - \frac{1}{4}z^{-1}}}_{H_2(z)} + \underbrace{\frac{-16 + 20z^{-1}}{1 - z^{-1} + \frac{1}{2}z^{-2}}}_{H_3(z)}$$



3) Sketch the structures for

$$H(z) = \frac{z(z-1)(z-2)(z+1)}{\left[z - \left(\frac{1}{2} + j\frac{1}{2}\right)\right] \left[z - \left(\frac{1}{2} - j\frac{1}{2}\right)\right] \left[z - \frac{j}{4}\right] \left[z + \frac{j}{4}\right]}$$

Soln:-

$$H(z) = \frac{(z^2 - z)(z^2 - z - 2)}{\left[\left(z - \frac{1}{2}\right)^2 + \left(\frac{1}{2}\right)^2\right] \left[z^2 + \left(\frac{1}{4}\right)^2\right]}$$

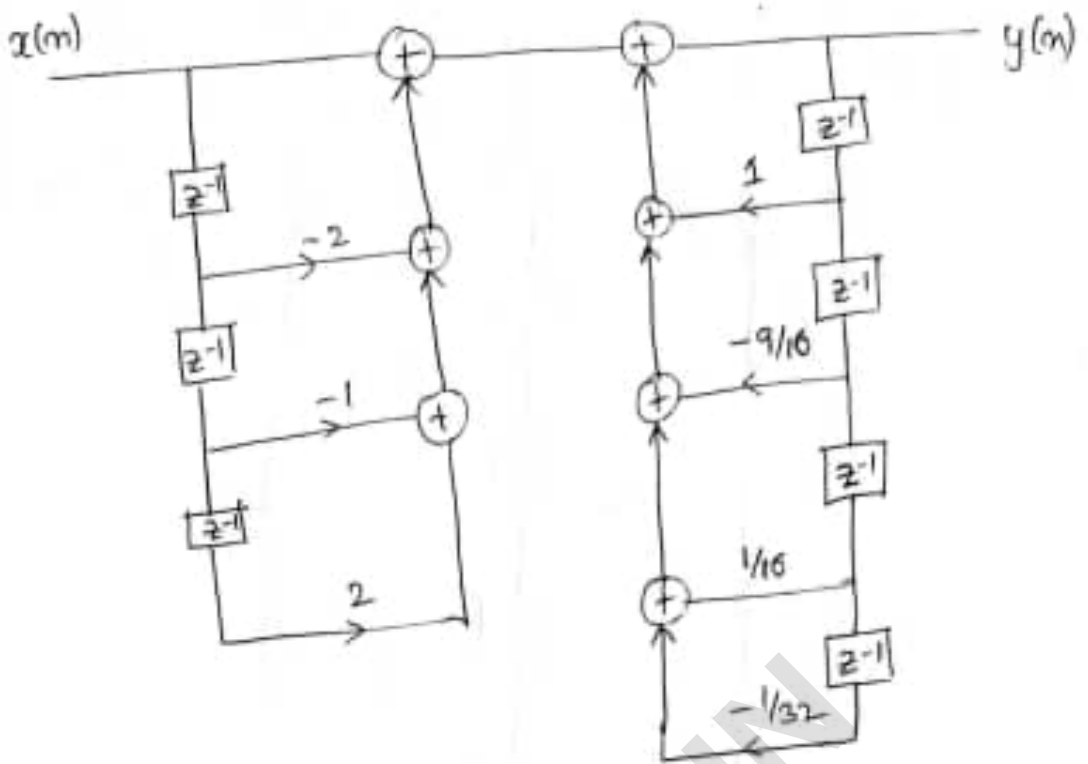
$$H(z) = \frac{z^4 - z^3 - 2z^2 - z^3 + z^2 + 2z}{\left[\left(z^2 - z + \frac{1}{4}\right) + \frac{1}{4}\right] \left[z^2 + \frac{1}{16}\right]}$$

$$H(z) = \frac{z^4 - 2z^3 - z^2 + 2z}{z^4 - z^3 + \frac{9}{16}z^2 - \frac{1}{16}z + \frac{1}{32}}$$

i) Direct Form-I:-

$$H(z) = \frac{Y(z)}{X(z)} = \frac{1 - 2z^{-1} - z^{-2} + 2z^{-3}}{1 - z^{-1} + \frac{9}{16}z^{-2} - \frac{1}{16}z^{-3} + \frac{1}{32}z^{-4}}$$

$$y(n) = x(n) - 2x(n-1) - x(n-2) + 2x(n-3) \\ + y(n-1) - \frac{9}{16}y(n-2) + \frac{1}{16}y(n-3) + \frac{1}{32}y(n-4)$$



(ii) Direct Form-II:-

$$\text{Let } H(z) = \frac{Y(z)}{X(z)} = \frac{V(z)}{V(z)} = \frac{Y(z)}{V(z)} \cdot \frac{V(z)}{X(z)}$$

$$\text{Let } \frac{Y(z)}{V(z)} = 1 - 2z^{-1} - z^{-2} + 2z^{-3}$$

Taking IZT,

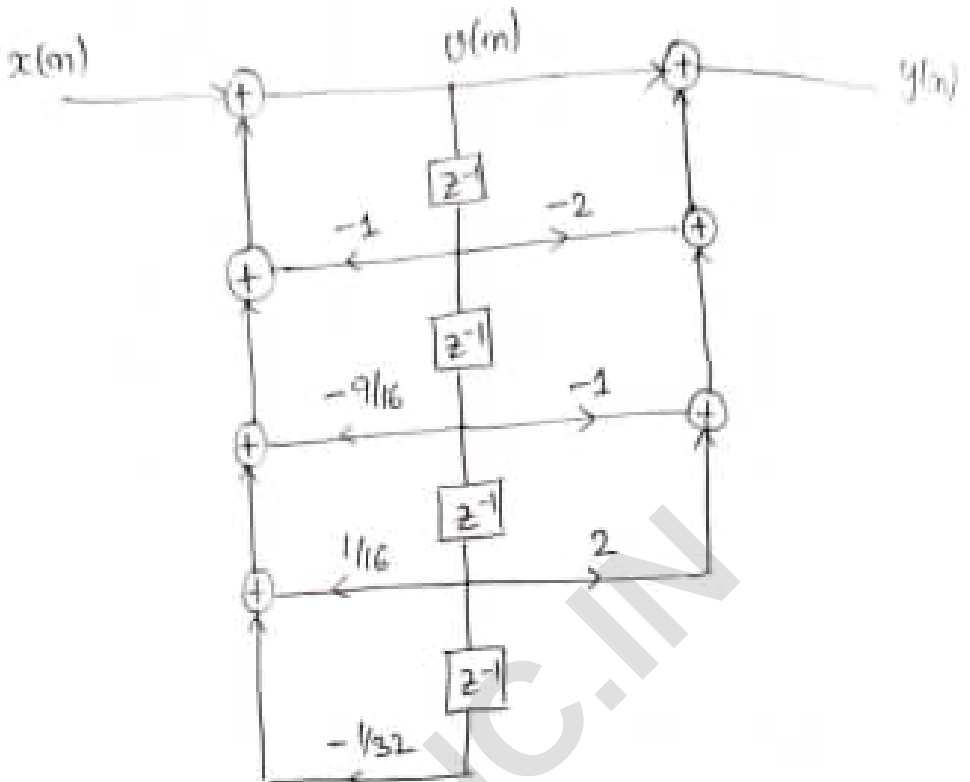
$$y(n) = v(n) - 2v(n-1) - v(n-2) + 2v(n-3) \quad \leftarrow (a)$$

$$\frac{V(z)}{X(z)} = \frac{1}{1 + z^{-1} + \frac{9}{16}z^{-2} - \frac{1}{16}z^{-3} + \frac{1}{32}z^{-4}}$$

$$\xrightarrow{\text{IZT}} x(n) = v(n) + v(n-1) + \frac{9}{16}v(n-2) - \frac{1}{16}v(n-3) + \frac{1}{32}v(n-4)$$

$$\therefore v(n) = x(n) - v(n-1) - \frac{9}{16}v(n-2) + \frac{1}{16}v(n-3) - \frac{1}{32}v(n-4) \quad \leftarrow (b)$$

From Eqns (a) & (b),



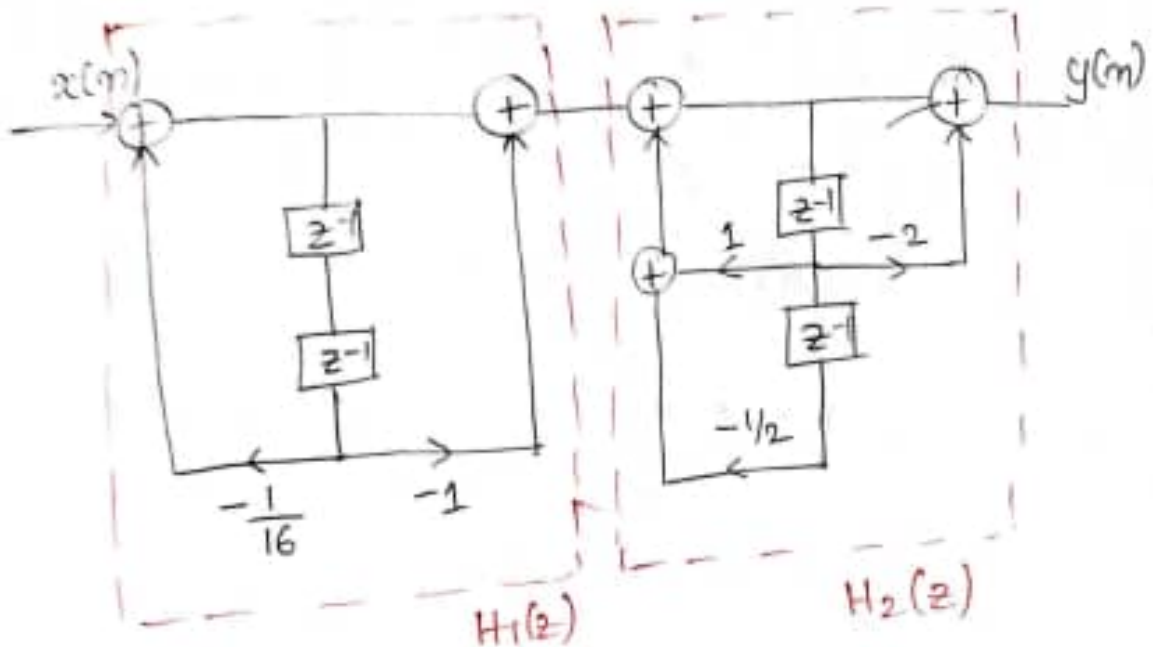
(iii) Cascade

$$H(z) = \frac{(z^2 - 8z)(z^2 - 1)}{(z^2 - z + 1/2)(z^2 + 1/16)}$$

$$H(z) = \frac{(1 - 8z^{-1})(1 - z^{-2})}{\left(1 + \frac{1}{16}z^{-2}\right)\left(1 - z^{-1} + \frac{1}{8}z^{-2}\right)}$$

$$\text{Let } H_1(z) = \left(\frac{1 - z^{-2}}{1 + \frac{1}{16}z^{-2}} \right) \times \left(\frac{1 - 8z^{-1}}{1 - z^{-1} + \frac{1}{8}z^{-2}} \right)$$

$$H(z) = H_1(z) \cdot H_2(z)$$



(m-1) → MEM LOCATIONS
 m → MULTIPLICATION
 Obtain Direct Form Realization for TRANSVERSAL OR TAPPED-DELAY LINE FILTER

$$H(z) = 1 + 2z^{-1} - 3z^{-2} - 4z^{-3} + 5z^{-4}$$

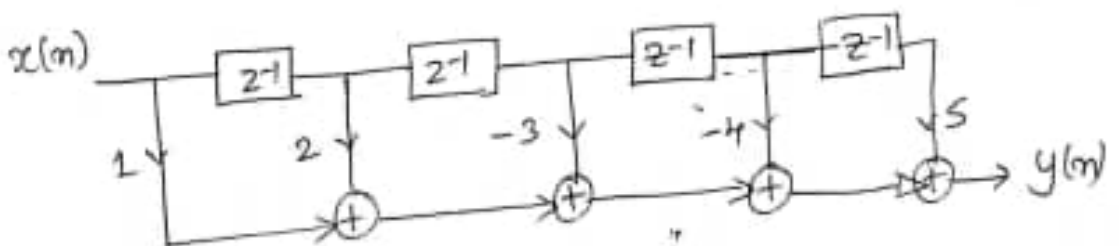
Soln:-

$$H(z) = \frac{Y(z)}{X(z)} = 1 + 2z^{-1} - 3z^{-2} - 4z^{-3} + 5z^{-4}$$

$$\text{or } Y(z) = X(z) + 2z^{-1}X(z) - 3z^{-2}X(z) - 4z^{-3}X(z) + 5z^{-4}X(z)$$

taking IZT,

$$y(n) = x(n) + 2x(n-1) - 3x(n-2) - 4x(n-3) + 5x(n-4)$$



DF-I REALIZATION